$$\because \quad x = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$\Rightarrow \quad x = \frac{1 \pm \sqrt{3}i}{2} \quad (\because \sqrt{-1} = i)$$

Thus, the three cube roots of unity are:

$1, \frac{-1+\sqrt{3}i}{2}$ and $\frac{-1+\sqrt{3}i}{2}$

Properties of Cube Roots of Unity

1. Each complex cube root of unity is square of the other

   If $\frac{-1+\sqrt{3}i}{2} = \omega$, then $\frac{-1-\sqrt{3}i}{2} = \omega^2$, and if $\frac{-1-\sqrt{3}i}{2} = \omega$, then $\frac{-1+\sqrt{3}i}{2} = \omega^2$ [ω is read as omega]
2. The sum of all the three cube roots of unity is zero i.e., $1 + \omega + \omega^2 = 0$
3. The product of all the three cube roots of unity is unity i.e., $1 \cdot \omega \cdot \omega^2 = \omega^3 = 1$

Four Fourth Roots of Unity

Let x be the fourth root of unity.

$$\because \quad x = (1)^{\frac{1}{4}}$$
$$\Rightarrow \quad x^4 = 1$$
$$\Rightarrow \quad x^4 - 1 = 0$$
$$\Rightarrow \quad (x^2-1)(x^2+1) = 0$$
$$\Rightarrow \quad x^2 - 1 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$
$$\text{and} \quad x^2 + 1 = 0 \Rightarrow x^2 = -1 \Rightarrow x = \pm i.$$

Hence four fourth roots of unity are: $1, -1, i, -i$.

Properties of four Fourth Roots of Unity

We have found that the four fourth roots of unity are: $1, -1, +i, -i$

1. Sum of all the four fourth roots of unity is zero

   $\because \quad 1 + (-1) + i + (-i) = 0$
2. The real fourth roots of unity are additive inverses of each other. 1 and -1 are the real fourth roots of unity and $1 + (-1) = 0 = (-1) + 1$
3. Both the imaginary fourth roots of unity are conjugate of each other

   i and $-i$ are imaginary fourth roots of unity, which are obviously conjugates of each other.
4. Product of all the fourth roots of unity is -1 i.e., $1 \times (-1) \times i \times (-i) = -1$

Example 1: Prove that:

$$(x^3 + y^3) = (x + y)(x + \omega y)(x + \omega^2 y)$$

Sol. R.H.S $= (x + y)(x + \omega y)(x + \omega^2 y)$

$= (x + y)[x + (\omega + \omega^2)yx + \omega^2 y^2]$

$= (x + y)(x^2 - xy + y^2) = x^3 + y^3 =$ L.H.S

$\{\because \omega^3 = 1, \omega + \omega^2 = -1\}$

Hence proved.

Note: We know that the numbers containing i are called imaginary number. So $\frac{-1+\sqrt{3}i}{2}$ and $\frac{-1-\sqrt{3}i}{2}$ are called complex or imaginary cube roots of unity.

Exercise 1.4

Q1. Find the three cube roots of;

(i) 8

Sol. Let x be a cube root of 8

$x = 8^{\frac{1}{3}}$

Take cube on both sides

$x^3 = \left(8^{\frac{1}{3}}\right)^3$

$x^3 = 8$

$x^3 - 8 = 0$

$x^3 - 2^3 = 0$

$(x-2)[(x)^2 + (x)(2) + (2)^2] = 0$

$(x-2)(x^2 + 2x + 4) = 0$

Either $x - 2 = 0$, $x = 2$

Or $x^2 + 2x + 4 = 0$

Here $a = 1, b = 2, c = 4$

By using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(4)}}{2(1)}$$

$$x = \frac{-2 \pm \sqrt{4-16}}{2}$$

$$x = \frac{-2 \pm \sqrt{-12}}{2}$$

$$x = \frac{-2 \pm 2\sqrt{3}i}{2}$$

$$x = \frac{-2 + 2\sqrt{3}i}{2} \text{ or } x = \frac{-2 - 2\sqrt{3}i}{2}$$

$$x=\frac{2\left(-1+\sqrt{3}i\right)}{2} \text{ or } x=\frac{2\left(-1-\sqrt{3}i\right)}{2}$$

$$x = 2\omega \text{ or } x = 2\omega^2$$

Solution set = $\{2, 2\omega, 2\omega^2\}$

(ii) -8

Sol. Let x be a cube root of -8

$$x=(-8)^{\frac{1}{3}}$$

Take cube on both sides

$$x^3=\left[(-8)^{\frac{1}{3}}\right]^3$$

$$x^3 = -8$$

$$x^3 + 8 = 0$$

$$(x+2)[(x)^2-(x)(2)+(2)^2]=0$$

$$(x+2)(x^2-2x+4)=0$$

Either $x+2=0$, $x=-2$

Or $x^2-2x+4=0$

Here $a=1$, $b=2$, $c=4$

By using quadratic formula

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$x=\frac{-(-2)\pm\sqrt{(-2)^2-4(1)(4)}}{2(1)}$$

$$x=\frac{2\pm\sqrt{4-16}}{2}$$

$$x=\frac{2\pm\sqrt{-12}}{2}$$

$$x=\frac{2\pm2\sqrt{3}i}{2}$$

$$x=\frac{2+2\sqrt{3}i}{2} \text{ or } x=\frac{2-2\sqrt{3}i}{2}$$

$$x=\frac{-2\left(-1-\sqrt{3}i\right)}{2} \text{ or } x=\frac{-2\left(-1+\sqrt{3}i\right)}{2}$$

$$x=-2\omega \text{ or } x=-2\omega^2$$

Solution set = $\{-2, -2\omega, -2\omega^2\}$

(iii) -27

Sol. Let x be a cube root of -27

$$x=(-27)^{\frac{1}{3}}$$

Take cube on both sides

$$x^3=\left[(-27)^{\frac{1}{3}}\right]^3$$

$$x^3=-27$$

$$x^3+27=0$$

$$x^3+3^3=0$$

$$(x+3)[(x)^2-(x)(3)+(3)^2]=0$$

$$(x+3)(x^2-3x+9)=0$$

Either $x+3=0$, $x=-3$

Or $x^2-3x+9=0$

Here a = 1, b = −3, c = 9

By using quadratic formula

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$x=\frac{-(-3)\pm\sqrt{(-3)^2-4(1)(9)}}{2(1)}$$

$$x=\frac{3\pm\sqrt{9-36}}{2}$$

$$x=\frac{3\pm\sqrt{-27}}{2}$$

$$x=\frac{3\pm3\sqrt{3}i}{2}$$

$$x=\frac{3+3\sqrt{3}i}{2} \text{ or } x=\frac{3-3\sqrt{3}i}{2}$$

$$x=\frac{-3\left(-1-\sqrt{3}i\right)}{2} \text{ or } x=\frac{-3\left(-1+\sqrt{3}i\right)}{2}$$

$$x=-3\omega \text{ or } x=-\omega^2$$

Solution set = $\{-3, 3\omega, -3\omega^2\}$

(iv) 64

Sol. Let x be a cube root of 64

$$x = 64$$

Take cube on both sides

$$x^3=\left(64^{\frac{1}{3}}\right)^3$$

$$x^3 = 64$$

$$x^3 = 4^3 = 0$$

$$(x-4)[(x)^2+(x)(4)+(4)^2]=0$$

$$(x-4)(x^2+4x+16)=0$$

Either $x-4=0$, $x=4$

Or $x^2+4x+16=0$

Here $a=1$, b = 4, $c=16$

By using quadratic formula

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$x=\frac{-4\pm\sqrt{4^2-4(1)(16)}}{2(1)}$$

Take power 4 on both sides

$x^4 = \left(625^{\frac{1}{4}}\right)^4$

$x^4 = 625$

$\Rightarrow\ x^4 - 625 = 0$,

$\Rightarrow\ (x^2)^2 - (25)^2 = 0$

$(x^2 + 25)(x^2 - 25) = 0$

$x^2 + 25 = 0$	$x^2 - 25 = 0$
$x^2 = -25$	$x^2 = 25$
$\sqrt{x^2} = \pm\sqrt{-25}$	$\sqrt{x^2} = \pm\sqrt{25}$
$x = \pm 5i$	$x = \pm 5$
$x = 5i, x = -5i$	$x = 5, x = -5$

Now, add them.

$= 5i + (-5i) + 5 + (-5)$

$= 5i - 5i + 5 - 5 = 0$

Hence, proved that their sum is zero.

Q3. If $1, \omega, \omega^2$ are the cube roots of unity, show that $1 + \omega^n + \omega^{2n} = 3$ where n is a multiple of 3 respectively.

Sol. Proof:

L.H.S $= 1 + \omega^n + \omega^{2n}$

L.H.S $= 1 + \omega^{3k} + \omega^{2(3k)}$

$\because$ n = multiple of 3 = $3k$ where $k \in z$

L.H.S $= 1 + \omega^{3k} + \omega^{6k}$

L.H.S $= 1 + (\omega^3)^k + (\omega^3)^{2k}$

L.H.S $= 1 + (1)^k + (1)^{2k}$ $\quad \because \omega^3 = 1$

L.H.S $= 1 + 1 + 1$

L.H.S $= 3$

L.H.S = R.H.S

Q4. Evaluate:

(i) $\left(\frac{-1+\sqrt{-3}}{2}\right)^7 + \left(\frac{-1-\sqrt{-3}}{2}\right)^7$

Sol. $\left(\frac{-1+\sqrt{-3}}{2}\right)^7 + \left(\frac{-1-\sqrt{-3}}{2}\right)^7$

We know that $\omega = \frac{-1+\sqrt{-3}}{2}$ and $\omega^2 = \frac{-1-\sqrt{-3}}{2}$

$= (\omega)^7 + (\omega^2)^7$

$= (\omega)^7 + (\omega)^{14}$

$= \omega^7 + \omega^{14}$

$= \omega.\omega^6 + \omega^2.\omega^{12}$

$= \omega.(\omega^3)^2 + \omega^2.(\omega^3)^4$

$= \omega.(1)^2 + \omega^2 (1)^4$

$= \omega + \omega^2 = -1$

(ii) $(-1+\sqrt{-3})^5 + (-1-\sqrt{-3})^5$

Sol. $\left(-1+\sqrt{-3}\right)^5 + \left(-1-\sqrt{-3}\right)^5$

We know that $\omega = \frac{-1+\sqrt{-3}}{2}$ and $\omega^2 = \frac{-1-\sqrt{-3}}{2}$

$\left(-1+\sqrt{-3}\right)^5 + \left(-1-\sqrt{-3}\right)^5$

$\Rightarrow\ = \left[\frac{2\left(-1+\sqrt{-3}\right)}{2}\right]^5 + \left[\frac{2\left(-1-\sqrt{-3}\right)}{2}\right]^5$

$= \left[2\left(\frac{-1+\sqrt{-3}}{2}\right)\right]^5 + \left[2\left(\frac{-1-\sqrt{-3}}{2}\right)\right]^5$

$= \left[2(\omega)\right]^5 + \left[2(\omega^2)\right]^5$

$= 2^5.\omega^5 + 2^5.(\omega^2)^5$

$= 2^5.\omega^{3+2} + 2^5.(\omega^{10})$

$= 32(1)\omega^2 + 32(1)\omega$

$= 32(\omega^2 + \omega)$ $\quad (\because \omega^3 = 1)$

$= 32(-1) = -32$

Q5. Show that $(1 - \omega + \omega^2)(1 - \omega^2 + \omega^4)(1 - \omega^4 + \omega^8)(1 - \omega^8 + \omega^{16})$... to 2n factors = 2^{2n}.

Sol. Proof:

L.H.S $= (1-\omega+\omega^2)(1-\omega^2+\omega^4)(1-\omega^4+\omega^8)(1-\omega^8+\omega^{16})$ .... to $2n$ factors

L.H.S $= (1-\omega+\omega^2)(1-\omega^2+\omega^3\omega)(1-\omega^3\omega+\omega^6\omega^2)$ $(1-\omega^6\omega^2+\omega^{15}\omega)$ .... to $2n$ factors

L.H.S $= \left(1-\omega+\omega^2\right)\left[1-\omega^2+(1)\omega\right]\left[1-(1)\omega+\omega^{3\times2}\omega^2\right]$ $\left(1-\omega^{3\times2}\omega^2+\omega^{3\times5}\omega\right)$...... to 2n factors

L.H.S $= \left(1-\omega+\omega^2\right)\left(1-\omega^2+\omega\right)\left[1-\omega+(1)^2\omega^2\right]$ $\left[1-(1)^2\omega^2+(1)^5\omega\right]$...... to 2n factors

L.H.S $= \left(1-\omega+\omega^2\right)\left(1+\omega-\omega^2\right)\left(1-\omega+\omega^2\right)$ $\left(1+\omega-\omega^2\right)$...... to 2n factors

L.H.S $= \left[\left(1-\omega+\omega^2\right)\left(1+\omega-\omega^2\right)\right]$ $\left[\left(1-\omega+\omega^2\right)\left(1+\omega-\omega^2\right)\right]$...... to n factors

L.H.S $= \left[1\cancel{+\omega}-\omega^2\cancel{-\omega}\cancel{-\omega^2}+\omega^3\cancel{+\omega^2}+\omega^3-\omega^4\right]$ $\left[1\cancel{+\omega}-\omega^2\cancel{-\omega}\cancel{-\omega^2}+\omega^3\cancel{+\omega^2}+\omega^3-\omega^4\right]$... to n factors

L.H.S $= \left[1-\omega^2+1+1-\omega^3\omega\right]\left[1-\omega^2+1+1-\omega^3\omega\right]$ ...... to n factors $\quad \because \omega^3 = 1$

$$L.H.S=\left[1-\omega^2+1+1-\omega\right]\left[1-\omega^2+1+1-\omega\right]$$
.... to n factors $\because \omega^3=1$

$L.H.S=\left(3-\omega-\omega^2\right)\left(3-\omega-\omega^2\right)$... to n factors

$L.H.S=\left[3-\left(\omega+\omega^2\right)\right]\left[3-\left(\omega+\omega^2\right)\right]$... to n factors

$L.H.S=[3-1][3-1]$ ....to n factors $\because 1=-\omega-\omega^2$

L.H.S = (2) (2).. to n factors

$L.H.S=2^n$

L.H.S = R.H.S

$\because$ L.H.S = R.H.S

Q6. **Prove that** $\left(\frac{i+\sqrt{3}}{2}\right)^8+\left(\frac{i-\sqrt{3}}{2}\right)^8=-1$

Sol. Proof:

$$L.H.S=\left(\frac{i+\sqrt{3}}{2}\right)^8+\left(\frac{i-\sqrt{3}}{2}\right)^8$$

$$L.H.S=\left(\frac{i\left(i+\sqrt{3}\right)}{2i}\right)^8+\left(\frac{i\left(i-\sqrt{3}\right)}{2i}\right)^8$$

$$L.H.S=\left(\frac{i^2+\sqrt{3}i}{2i}\right)^8+\left(\frac{i^2-\sqrt{3}i}{2i}\right)^8$$

$$L.H.S=\left(\frac{-1+\sqrt{3}i}{2i}\right)^8+\left(\frac{-1-\sqrt{3}i}{2i}\right)^8$$

$$L.H.S=\left[\frac{1}{i}\left(\frac{-1+\sqrt{3}i}{2}\right)\right]^8+\left[\frac{1}{i}\left(\frac{-1-\sqrt{3}i}{2}\right)\right]^8$$

$$L.H.S=\left(\frac{1}{i}\right)^8\left(\frac{-1+\sqrt{3}i}{2}\right)^8+\left(\frac{1}{i}\right)^8\left(\frac{-1-\sqrt{3}i}{2}\right)^8$$

$$L.H.S=\frac{1}{i^8}\left[\left(\frac{-1+\sqrt{3}i}{2}\right)^8+\left(\frac{-1-\sqrt{3}i}{2}\right)^8\right]$$

$$L.H.S=\frac{1}{i^{2\times4}}\left[(\omega)^8+\left(\omega^2\right)^8\right]$$

$$L.H.S=\frac{1}{(-1)^4}\left(\omega^8+\omega^{16}\right)$$

$$L.H.S=\frac{1}{1}\left(\omega^6\omega^2+\omega^{15}\omega\right)$$

$L.H.S=\omega^2+\omega$ $\because \omega^6=\omega^{15}=1$

$L.H.S=-1$ $\because \omega+\omega^2=-1$

L.H.S = R.H.S

Q7. **Evaluate** $\sum_{k=0}^{5}\omega^{2k}$ **, where ω is an imaginary cube root of unity.**

Sol. $\sum_{k=0}^{5}\omega^{2k}$

$=\omega^{2(0)}+\omega^{2(1)}+\omega^{2(2)}+\omega^{2(3)}+\omega^{2(4)}+\omega^{2(5)}$

$=\omega^0+\omega^2+\omega^4+\omega^6+\omega^8+\omega^{10}$

$=1+\omega^2+\omega^3\omega+\omega^6+\omega^6\omega^2+\omega^9\omega$

$=1+\omega^2+\omega+1+\omega^2+\omega$ $\because \omega^3=1$

$=0+0=0$ $\because 1+\omega+\omega^2=0$

Q8. **If ω is an imaginary cube root of unity, prove that** $\frac{a+b\omega^2+c\omega}{a\omega^2+b\omega+c}=\omega$.

Sol. Proof:

$$L.H.S=\frac{a+b\omega^2+c\omega}{a\omega^2+b\omega+c}$$

$$L.H.S=\frac{\omega^2\left(a+b\omega^2+c\omega\right)}{\omega^2\left(a\omega^2+b\omega+c\right)}$$

$$L.H.S=\frac{a\omega^2+b\omega^4+c\omega^3}{\omega^2\left(a\omega^2+b\omega+c\right)}$$

$$L.H.S=\frac{a\omega^2+b\omega^3\omega+c\omega^3}{\omega^2\left(a\omega^2+b\omega+c\right)}$$

$$L.H.S=\frac{a\omega^2+b\omega+c}{\omega^2\left(a\omega^2+b\omega+c\right)} \qquad \because \omega^3=1$$

$$L.H.S=\frac{1}{\omega^2}$$

$$L.H.S=\frac{1\omega}{\omega^2\omega}$$

$$L.H.S=\frac{\omega}{\omega^3}$$

$$L.H.S=\frac{\omega}{1}$$

$L.H.S=\omega$

L.H.S = R.H.S

Q9. **If ω is a cube root of unity, prove that** $\frac{a\omega^{12}+b\omega^{17}+c\omega^{19}}{a\omega^{14}+b\omega^{22}+c\omega^{30}}=\omega$

Sol. Proof:

$$L.H.S=\frac{a\omega^{12}+b\omega^{17}+c\omega^{19}}{a\omega^{14}+b\omega^{22}+c\omega^{30}}$$

$$L.H.S=\frac{a\omega^{12}+b\omega^{15}\omega^2+c\omega^{18}\omega}{a\omega^{12}\omega^2+b\omega^{21}\omega+c\omega^{30}}$$

$$L.H.S = \frac{a+b\omega^2+c\omega}{a\omega^2+b\omega+c} \qquad \because \omega^{3^k}=1$$

$$L.H.S = \frac{\omega^2\left(a+b\omega^2+c\omega\right)}{\omega^2\left(a\omega^2+b\omega+c\right)}$$

$$L.H.S = \frac{a\omega^2+b\omega^4+c\omega^3}{\omega^2\left(a\omega^2+b\omega+c\right)}$$

$$L.H.S = \frac{a\omega^2+b\omega^3\omega+c\omega^3}{\omega^2\left(a\omega^2+b\omega+c\right)}$$

$$L.H.S = \frac{a\omega^2+b\omega+c}{\omega^2\left(a\omega^2+b\omega+c\right)} \qquad \because \omega^3=1$$

$$L.H.S = \frac{1}{\omega^2}$$

$$L.H.S = \frac{1\omega}{\omega^2\omega}$$

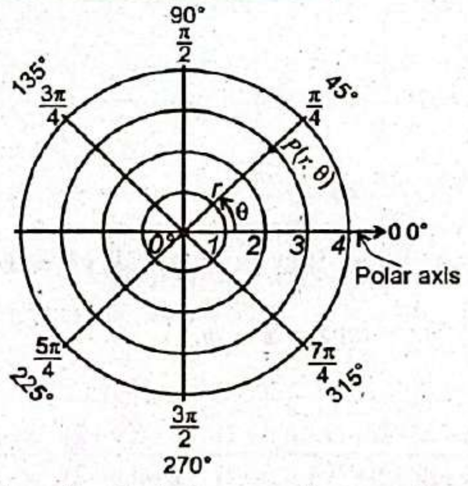
$$L.H.S = \frac{\omega}{\omega^3}$$

$$L.H.S = \frac{\omega}{1}$$

$$L.H.S = \omega$$

$$L.H.S = R.H.S$$

1.5 Polar Coordinates System



Just as the Cartesian coordinate system uses an orderd pair (x, y) to describe the position of a point, the polar coordinate system determines the position of a point using a directed distance r from a fixed origin O (called the pole) and an angle θ theat the line connecting the origin to the point makes with the polar axis (typically aligned with the positive x-axis).

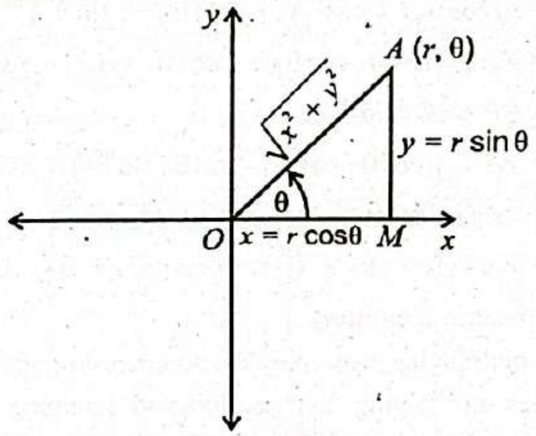
In polar coordinate system the location of a point P can be described by polar coordinates in the form (r, θ), where r and θ are real numbers.

Consider the adjoining diagram representing the complex number $z = x + iy$. From the diagram, we see that $x = r\cos\theta$ and $y = r\sin\theta$, where $r = |z|$ is modulus and θ is called an argument of.

Hence $x + iy = r\cos\theta + i\,r\sin\theta$ ......(i)

where $r = |z| = \sqrt{x^2+y^2}$ and $\theta = tan^{-1}\frac{y}{z}$ $(x \neq 0)$

Equation (i) is called the polar form of the complex number z.



Example 1: Express the complex number $1+i\sqrt{3}$ in polar form.

Sol. Step-1: Put $r\cos\theta = 1$ and $r\sin\theta = \sqrt{3}$

Step - II: $r^2 = (1)^2 + \left(\sqrt{3}\right)^2$

$\Rightarrow \quad r^2 = 1 + 3 = 4$

$\Rightarrow \quad r = 2$

Step - III: $\theta = tan^{-1}\frac{\sqrt{3}}{1} = tan^{-1}\sqrt{3} = 60°$

Thus $1 + i\sqrt{3} = 2\cos 60° + i\,2\sin 60°$

Principal Argument: The principal argument θ of a complex number $z = a + bi$ is the angle between the positive real axis and the line joining (a,b) to the origin in the Argand plane.

$$Arg\ z = \theta = tan^{-1}\left(\frac{b}{a}\right),\ (a \neq 0)$$

It is denoted by Arg. It is a single, specific value of the argument, typically chosen within a standard range: Arg $z \in (-\pi, \pi]$.

Operations on Complex Numbers in Polar Form

Addition and Subtraction of Complex number in Polar form:

Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ be two complex number in polar form. The addition and subtraction of two numbers can be computed simply as: